Chapter 1: Similarity

Q1. In ΔABC, DE ∥ BC. If AD = 3 cm, DB = 6 cm and EC = 8 cm, find AE.
Given: In ΔABC, D lies on AB and E lies on AC such that DE ∥ BC. AD = 3 cm, DB = 6 cm, EC = 8 cm.
To find: AE
Step 1: By Basic Proportionality Theorem (BPT),
AD/DB = AE/EC
Substitute: 3/6 = AE/8
⇒ 1/2 = AE/8
⇒ AE = 8/2 = 4 cm
Answer: AE = 4 cm.

Q2. In ΔPQR, XY ∥ QR. If PX = 4 cm, XQ = 6 cm and PR = 15 cm, find PY.
Given: In ΔPQR, X lies on PQ and Y lies on PR, with XY ∥ QR. PX = 4 cm, XQ = 6 cm, PR = 15 cm.
To find: PY
Step 1: PQ = PX + XQ = 4 + 6 = 10 cm.
Step 2: By BPT, PX/PQ = PY/PR
⇒ 4/10 = PY/15
⇒ 2/5 = PY/15
⇒ PY = 15 × (2/5) = 6 cm
Answer: PY = 6 cm.

Q3. If ΔABC ~ΔPQR and AB = 6 cm, PQ = 9 cm, BC = 8 cm, find QR.
Given: ΔABC ~ΔPQR, AB = 6 cm, PQ = 9 cm, BC = 8 cm.
To find: QR
Since triangles are similar, corresponding sides are proportional:
AB/PQ = BC/QR
⇒ 6/9 = 8/QR
⇒ 2/3 = 8/QR
⇒ QR = 8 × (3/2) = 12 cm
Answer: QR = 12 cm.

Q4. If area(ΔABC) : area(ΔPQR) = 16 : 25, find AB : PQ.
Given: area(ΔABC) : area(ΔPQR) = 16 : 25
To find: AB : PQ
For similar triangles, ratio of areas = square of ratio of corresponding sides:
area(ΔABC)/area(ΔPQR) = $(AB/PQ)^2$
⇒ 16/25 = $(AB/PQ)^2$
Taking square root on both sides: AB/PQ = 4/5
Answer: AB : PQ = 4 : 5.

Q5. A line parallel to one side of a triangle divides other sides in ratio 3 : 5. If one segment is 6 cm, find the other.
Given: A line ∥ one side divides the other two sides in the ratio 3:5. One segment = 6 cm.
To find: The other segment
Let the two corresponding segments be 3x and 5x.
Given 3x = 6 ⇒ x = 2
Other segment = 5x = 5×2 = 10 cm.
Answer: The other segment is 10 cm.

Q6. In ΔABC, PQ ∥ BC. If area(APQ) : area(ABC) = 1 : 4, find AP : AB.
Given: In ΔABC, PQ ∥ BC. area(ΔAPQ) : area(ΔABC) = 1 : 4
To find: AP : AB
Since PQ ∥ BC, ΔAPQ ~ΔABC.
For similar triangles: area ratio = square of corresponding side ratio:
area(APQ)/area(ABC) = $(AP/AB)^2$
⇒ 1/4 = $(AP/AB)^2$
⇒ AP/AB = 1/2
Answer: AP : AB = 1 : 2.

Q7. If ratio of perimeters of two similar triangles is 5 : 7, find ratio of their areas.
Given: Perimeter ratio of two similar triangles = 5 : 7
To find: Area ratio
In similar triangles, ratio of perimeters = ratio of corresponding sides.
Thus side ratio = 5 : 7
Area ratio = $(\text{side ratio})^2 = 5^2 : 7^2 = 25 : 49$
Answer: Ratio of areas = 25 : 49.

Q8. Prove that a line drawn parallel to one side of a triangle divides other two sides proportionally. (Basic Proportionality Theorem)
Given: In ΔABC, a line DE is drawn such that D lies on AB, E lies on AC and DE ∥ BC.
To prove: AD/DB = AE/EC
Proof:
1) Since DE ∥ BC, we have ∠ADE = ∠ABC and ∠AED = ∠ACB (corresponding angles).
2) Also ∠A is common in ΔADE and ΔABC.

3) Therefore, ΔADE ~ΔABC (AA similarity).
4) Hence, corresponding sides are proportional: AD/AB = AE/AC.
5) Write AB = AD + DB and AC = AE + EC.
From AD/AB = AE/AC ⇒ AD/(AD+DB) = AE/(AE+EC).
Cross-multiplying: AD(AE+EC) = AE(AD+DB)
⇒ ADAE + ADEC = ADAE + AE·DB
⇒ AD·EC = AE·DB
⇒ AD/DB = AE/EC.
Hence proved.

Q9. Prove the converse of Basic Proportionality Theorem.
Given: In ΔABC, D lies on AB and E lies on AC such that AD/DB = AE/EC.
To prove: DE ∥ BC
Proof:
1) Through point D, draw a line DE' ∥ BC meeting AC at E'.
2) By BPT (proved earlier) in ΔABC with DE' ∥ BC: AD/DB = AE'/E'C.
3) Given AD/DB = AE/EC.
4) Therefore AE'/E'C = AE/EC.
5) On the same line AC, a point dividing AC in a given ratio is unique. Hence E' coincides with E.
6) Therefore the line through D parallel to BC meets AC at E itself, i.e., DE ∥ BC.
Hence proved.

Q10. In ΔABC, AB = 12 cm, AD = 4 cm, AC = 15 cm, DE ∥ BC. Find AE.
Given: In ΔABC, D lies on AB, E lies on AC, DE ∥ BC. AB = 12 cm, AD = 4 cm, AC = 15 cm.
To find: AE
By BPT: AD/AB = AE/AC
⇒ 4/12 = AE/15
⇒ 1/3 = AE/15
⇒ AE = 15/3 = 5 cm
Answer: AE = 5 cm.